Orbita - IQ — Technical & Architecture Report

LIVE SYSTEM
orbita-iq.vercel.app

Ground Operations · Conjunction Intelligence Dashboard

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1. Project Overview

Orbita - IQ is a satellite operations and conjunction-intelligence dashboard built for a ground-ops-style workflow: track an operational fleet of satellites, screen them for close approaches against both each other and the broader tracked space object catalog, and surface actionable risk alerts to an operator — modeled loosely on real-world Conjunction Data Message (CDM) screening as performed by organizations such as the 18th Space Defense Squadron.

The system covers the full operational loop: **catalog ingestion** → **fleet tracking** → **orbit propagation** → **conjunction screening** → **risk classification** → **alerting** → **operator review**, complemented by a CesiumJS 3D visualization digital twin and an LLM-assisted advisory tool for qualitative conjunction assessment.

2. Tech Stack

Layer	Technology & Implementations
Frontend	React 18, Vite, TypeScript, Tailwind CSS, Lucide Icons
Frontend Hosting	Vercel (Static Single Page Application)
Backend	Python 3.11, FastAPI, SQLAlchemy (async, asyncpg driver)
Backend Hosting	Render (Managed Web Service)
Database / Auth	Supabase (Managed PostgreSQL, GoTrue Auth, Realtime Subscriptions, RLS)
Orbit Propagation	SGP4 / SDP4 (via python-sgp4, vectorized with NumPy arrays)
Scheduling	APScheduler (AsyncIOScheduler background tasks)
Numerics / TCA	NumPy, SciPy (scipy.optimize.minimize_scalar for TCA refinement)
3D Digital Twin	CesiumJS (Cartesian3 transformations, orbital paths, camera track)
Live Updates	Supabase Realtime + dedicated /ws/orbit WebSocket with 20s polling fallback
External Data	CelesTrak (Automated TLE / space catalog ingestion)

Note: Frontend and backend are deployed and scaled independently — Vercel serves the static SPA build, and all client calls communicate directly with the Render FastAPI service.

3. Data Model

Core tables in the Supabase PostgreSQL database:

- **satellites** — The operator's tracked fleet ('My Satellites'). Stores NORAD ID, name, international designator, object type, status, owning organization, and orbital state.
- **catalog_satellites** — Global space catalog (~5,000+ objects spanning payloads, debris, and rocket bodies across LEO/MEO/GEO/HEO), searchable and filterable.
- **tle_records** — Stored two-line element sets backing SGP4 propagation for every tracked and cataloged object.

- **orbit_state** — Current propagated position/velocity per satellite, refreshed on schedule and pushed to clients in real time.
- **conjunction_alerts / conjunction_events** — Single source of truth for screening results: satellite pair, scope, time of closest approach (TCA), miss distance, relative velocity, collision probability (Pc), risk level, and alert status.
- **profiles** — Operator identity (employee ID, full name, role, department), backed by Supabase Auth and strict RLS policies.

Schema Integrity: Several fields use native Postgres enum types (satellite_status, object_type, risk_level, alert_status, alert_state), strictly enforced at database DDL level.

4. Core Features

- **Dashboard** — Fleet-wide summary KPI cards (tracked count, active alerts, high-risk alerts, next upcoming conjunction), fleet altitude trend charts, and real-time live conjunction activity feed.
- **My Satellites** — Operator's active fleet (611 satellites in current dataset), each with live altitude, lat/long coordinates, velocity, and health status, refreshed via scheduled SGP4 propagation.
- **All Satellites (Catalog)** — Global catalog (~5,028 objects), filterable by orbital regime (LEO/MEO/GEO/HEO) and type (Payloads, Debris, Rocket Bodies), searchable by NORAD/COSPAR ID, with 'Sync Catalog' & 1-click 'Track' capabilities.
- **Alerts (CDM Screening)** — Conjunction Data Message-style screening results sorted by TCA, filterable by risk level (Critical/High/Medium/Low) and scope (Fleet vs Fleet, Fleet vs Catalog), with full vector detail modals & operator status workflow (Monitor/Resolve).
- **Orbit Viewer (3D Twin)** — CesiumJS-based 3D digital twin rendering live satellite positions and full orbital tracks, offering imagery layer switching, time-warp simulation controls (1x–60x), and live/global fleet toggles.
- **AI Assistant** — LLM-driven qualitative conjunction analysis and maneuver advisory generator, designed as an operational decision-support layer with cached advisories per conjunction event.
- **Settings** — Operator profile preferences, notification alert thresholds, and security controls.

5. Conjunction Screening Engine

The screening engine is the computational core of Orbita - IQ. It operates across two screening scopes — **Fleet vs. Fleet** and **Fleet vs. Catalog** — over a rolling **5-day (120-hour)** look-ahead horizon using a high-performance two-stage pipeline:

- **Stage 1 — Coarse Orbital Filter:** For every candidate pair, perigee/apogee altitude envelopes (with safety margins) are checked for overlap. Pairs whose orbital shells geometrically cannot intersect are discarded before any propagation. This eliminates **>99%** of raw pairs at virtually zero computational cost.
- **Stage 2 — Vectorized Ephemeris Propagation:** For all surviving unique satellites, a 5-day state trajectory is precomputed once using vectorized SGP4. Relative distance between pairs across the entire 120-hour timeline is evaluated in a single NumPy array pass. When a local minimum is detected, the true Time of Closest Approach (TCA) is refined via scipy.optimize.minimize_scalar. Per-pair scan cost is reduced to sub-millisecond speeds.

Risk Classification Thresholds

Miss Distance	Risk Level	Collision Probability (P_c)	Operator Action / Notification
< 1.0 km	CRITICAL	Evaluated per event	**Active Urgent Alert** + Banner
1.0 – 5.0 km	HIGH	Evaluated per event	**Active Alert** (Top Priority)
5.0 – 25.0 km	MEDIUM	Evaluated per event	Stored & Visible in Alerts table
25.0 – 50.0 km	LOW	Evaluated per event	Stored & Visible for situational awareness
> 50.0 km	Not Reported	N/A	Filtered out (noise suppression)

Scheduled Engine Automation:

- **Orbit Propagation (5 min):** Computes real-time satellite state vectors and updates orbit_state table.
- **Conjunction Screening (20 min):** Executes Stage 1 & Stage 2 screening across Fleet-vs-Fleet and Fleet-vs-Catalog.
- **Catalog Synchronization (12 hours):** Fetches fresh TLE ephemerides from CelesTrak to prevent propagation drift.

6. Real-Time Data Flow & WebSocket Architecture

```
                      +---------------------+
                      |  conjunction_alerts  |  (Single Source of Truth)
                      +----------+----------+
                                 |
         +-----------------------+-----------------------+
         |                       |                       |
         v                       v                       v
GET /api/v1/dashboard   GET /api/v1/alerts       Supabase Realtime
 - Active Alerts         - Filtered/sorted rows    - Live push to clients
 - High Risk Alerts      - Status updates          - 20s polling fallback
 - Next Conjunction      - Horizon / TCA badges
```

Live orbit positions follow a dedicated path: the scheduled propagation job writes updated coordinates to orbit_state and broadcasts over the /ws/orbit WebSocket channel, feeding the My Satellites table and CesiumJS 3D globe simultaneously.

7. Deployment Architecture

```
Vercel (React/Vite SPA)  ==== HTTP ====>  Render (FastAPI)  ==== asyncpg ====>  Supabase (PostgreSQL)
                                              |                                       |
                                              +-- APScheduler background jobs         +-- Supabase Auth
                                              +-- /ws/orbit WebSocket stream          +-- Realtime Push

CelesTrak  === (Catalog Sync Job) ===>  Render (FastAPI)  ========>  Supabase (catalog_satellites)
```

8. Notable Engineering Considerations

- **Schema / Migration Discipline:** Native PostgreSQL enum types stay strictly synchronized between SQLAlchemy data models and database DDL through structured Alembic migrations, failing loudly on schema drift rather than attempting risky runtime auto-mutations.
- **SGP4 Accuracy Over 5-Day Horizons:** Numerical propagation error accumulates over time elapsed since TLE epoch. A high-frequency 12-hour CelesTrak TLE sync cadence maintains positional fidelity.
- **Alert Fatigue Suppression:** With global screening encompassing both fleet-internal and catalog objects, lower-risk approaches vastly outnumber critical events. The system strictly isolates 'stored/cataloged data' from 'active high-risk notifications' to keep operator workflows focused and actionable.

9. Next Steps & Technical Roadmap

- **Blended Probability Classification:** Integrate collision probability (P_c) calculations directly into the primary risk tiering metric alongside miss distance.
- **Time-Series Conjunction Trends:** Transition dashboard altitude/conjunction trend graphs from static views to persistent historical time-series analytics.
- **Covariance-Based Conjunction Assessment:** Incorporate full state covariance matrices for high-precision probability volumes beyond standard TLE-derived point state vectors.